

Multi-AGN Monte Carlo Jet Power Batch

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PAPER_926: Multi-AGN Monte Carlo Jet Power Batch

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Source: SCm phonon gap implementation (agn_jet_power_curves.py)

Calculator: MultiAGNJetPowerMonteCarloBatchCalc (CP4 #510)

CVW: v2.0.0 compliant

Abstract

Monte Carlo sampling (10^6 samples) of phonon-enhanced jet power P_{jet} across four AGN archetypes: 3C273-type ($M = 8.86e8 M_{\text{sun}}$), CenA-type ($5.5e7 M_{\text{sun}}$), TON618-type ($6.6e10 M_{\text{sun}}$), and TXS0506-type ($3e8 M_{\text{sun}}$). Gamma draws from $N(\mu_{\text{Gamma}}, \sigma_{\text{Gamma}}^2)$ with $\mu = 1.25$ THz, $\sigma = 0.15$ THz to model thermal phonon linewidth fluctuations. Reports mean, standard deviation, median, p5, and p95 percentiles of $P_{\text{jet}}(\text{Gamma})$ for each system. The P_{BZ} baseline scales as M^2 , giving a dynamic range of $\sim 10^8$ in base jet power across the four systems.

1. Core Equations

Section A: Lagrangian

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$$
\begin{aligned}
& \& P_{\text{jet}} = P_{\text{BZ}} (1 + M_{\text{jet}}(\text{Gamma})) \parallel \\
& \& M_{\text{jet}}(\text{Gamma}) = 1 + A_{\text{jet}} \exp[-(\text{Gamma} - \text{Gamma}_0)^2 / (2\sigma_{\text{Gamma}}^2)] \parallel \\
& \& P_{\text{BZ}} = (\pi / (6\mu_0)) B^2 r_g^2 c a^2 \parallel \\
& \& \text{Gamma} \sim N(1.25 \text{ THz}, 0.15^2 \text{ THz}^2) \\
& \end{aligned}
$$
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Section B: VDS/DVP/BH Number Systems

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MC statistics: mean( $P_{\text{jet}}$ ), std( $P_{\text{jet}}$ ), median( $P_{\text{jet}}$ ),  $P_5$ ,  $P_{95}$ 
Modulation range: [min( $1+M_{\text{jet}}$ ), max( $1+M_{\text{jet}}$ )] over all draws
System diversity:  $P_{\text{BZ}}(\text{TON618}) / P_{\text{BZ}}(\text{CenA}) \sim (6.6e10/5.5e7)^2 \sim 1.4e6$ 
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Section SM: SM Anchors

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$$
\begin{aligned}
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& 3C273-type: M = 8.86e8 M_sun, high-luminosity quasar \\
& CenA-type: M = 5.5e7 M_sun, nearby radio galaxy \\
& TON618-type: M = 6.6e10 M_sun, ultramassive quasar \\
& TXS0506-type: M = 3e8 M_sun, blazar neutrino source
\end{aligned}
$$

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2. Parameters

Parameter	Default	Description
M_bh_Msun	8.86e8	BH mass (solar masses)
a_spin	0.9	Spin parameter
B_field	50 T	Magnetic field
A_jet	1.5	Modulation amplitude
sigma_Gamma_THz	0.08	sigma_Gamma
Gamma_mean_THz	1.25	MC mean Gamma
Gamma_std_THz	0.15	MC std dev
n_samples	100000	MC sample count

3. Key Results

System	P_BZ (erg/s)	<P_jet> (erg/s)	Spread
3C273-type	$\sim 10^{46}$	$\sim 2.5 P_{BZ}$	$\pm 30\%$
CenA-type	$\sim 10^{42}$	$\sim 2.5 P_{BZ}$	$\pm 30\%$
TON618-type	$\sim 10^{50}$	$\sim 2.5 P_{BZ}$	$\pm 30\%$
TXS0506-type	$\sim 10^{44}$	$\sim 2.5 P_{BZ}$	$\pm 30\%$

4. Physical Interpretation

The MC approach captures the stochastic nature of phonon linewidth fluctuations in real AGN environments. The 30% spread in P_{jet} at fixed BH mass and spin arises entirely from phonon frequency variations, providing a natural explanation for AGN jet power variability on timescales of hours to days (matching observed VHE flaring). The consistent $\sim 2.5x$ mean enhancement across all four systems demonstrates that phonon modulation is mass-independent, affecting only the multiplicative factor while P_{BZ} provides the mass-dependent baseline.

5. References

- PAPER_920: Monte Carlo jet power sampling (Session 210c)
- PAPER_925: Quasar jet phonon modulation $M_{jet}(\Gamma)$
- agn_jet_power_curves.py: 3-class standalone module

<!-- PKG-AGN-S225 -->

Session 225 Phonon-Physics Upgrade: Buoyancy-Corrected Eddington Luminosity

> Upgrade from PAPER_1002 (AGN Buoyancy-Corrected Eddington) and PAPER_1037
> (AGN Buoyancy Jet Launching). See also PAPER_1009-1010 for $F_{U_{Bi}}$ jet
> modulation curves and PAPER_1048 for phonon-corrected M - σ relation.

The SCm vacuum buoyancy partially opposes gravitational radiation pressure, raising the effective Eddington luminosity:

$$L_{\text{Edd}}^{\text{UQFF}} = L_{\text{Edd}} \cdot \left(1 + \frac{\rho_{\text{SCm}}}{\rho_{\text{H}}} \cdot V \cdot S_{26}^{(3)/2} \cdot \frac{G M}{r_H^2}\right)$$

where:

- L_{Edd} is the classical Eddington luminosity
- $\rho_{\text{SCm}} = 7.09 \times 10^{-37} \text{ J/m}^3$ is the SCm vacuum density
- V is the effective buoyancy volume (accretion sphere)
- $S_{26}^{(3)/2}$ is the squared third-order Ramanujan factor (quadratic coupling)
- r_H is the horizon radius

Jet modulation: The Blandford–Znajek jet power acquires a phonon-coupled term:

$$P_{\text{jet}}^{\text{UQFF}} = P_{\text{BZ}} \cdot \left[1 + \beta_i \cdot \Phi_{1.25}(\text{THz}) \cdot \left(\frac{B}{B_{\text{crit}}}\right)^2\right]$$

where $\Phi_{1.25}(\text{THz}) = \cos(\omega_{\text{SCm}} \cdot t)$ modulates jet power at the phonon frequency.

M - σ correction (PAPER_1048): The phonon-corrected M - σ relation becomes $M_{\text{BH}} \propto \sigma^{4+\delta}$ where $\delta = \beta_i \cdot S_{26}^{(3)/2} \cdot (\omega_{\text{SCm}} / \omega_{\text{bulge}})$.

<!-- PKG-LAG-S225 -->

Session 225 Phonon-Physics Upgrade: UQFF 9-Sector Lagrangian

> Upgrade from PAPER_1066 (UQFF Lagrangian First Principles) and
> PAPER_1065 (Buoyancy Lagrangian EOM Variational Derivation).

The complete UQFF Lagrangian density, from which all sector-specific equations of motion derive:

$$\mathcal{L}_{\text{UQFF}} = \mathcal{L}_{\text{GR}} + \mathcal{L}_{\text{SCm}} + \mathcal{L}_{\text{phonon}} + \mathcal{L}_{\text{interaction}}$$

$$\mathcal{L}_{\text{SCm}} = \frac{1}{2} (\partial_\mu \phi)^2 - \lambda |\phi|^2 - v_{\text{SCm}}^2 |\phi|^2$$

The SCm condensate potential minimum gives $V(\phi_0) = -7.09 \times 10^{-37} \text{ J/m}^3$ (matching ρ_{SCm}) and phonon mass $m_{\text{phonon}} = \sqrt{8\lambda} \cdot v_{\text{SCm}}$.

Nine-sector closure (Session 202):

$$\mathcal{L}_9 = \mathcal{L}_{\text{EH}} + \mathcal{L}_{\text{YM}} + \mathcal{L}_{\text{Dirac}} +$$

$\mathcal{L}_{\text{SCm}} + \mathcal{L}_{\text{mag}} + \mathcal{L}_{\text{buoy}} + \mathcal{L}_{\text{aether}} + \mathcal{L}_{\text{LENR}} + \mathcal{L}_{\text{KK}} \} \}$

Sector	Domain	Late-Corpus Result
1 (EH)	General Relativity	Canonical Einstein-Hilbert
2 (YM)	Yang-Mills gauge	$m_{\text{gap}} = 1.736 \text{ GeV}$ (PAPER_1318)
3 (Dirac)	Fermion / LENR	Kozima neutron-drop (PAPER_1061)
4 (SCm)	Superconducting manifold	$V(\phi_0) = -\rho_{\text{SCm}}$ canonical
5 (Mag)	Um magnetism	Heaviside amplifier (PAPER_1072)
6 (Buoy)	F_U_Bi_i buoyancy	Variational EOM (PAPER_1065)
7 (Aether)	Vacuum background	Two-component ρ (PAPER_1051)
8 (LENR)	Nuclear transmutation	COP parametric (PAPER_1081)
9 (KK)	Kaluza-Klein 26D	$S_{26}^{(3)}$ compactification (PAPER_1080)

Calibration Constants

Constant	Symbol	Value	Validation Domain
UQFF damping rate	κ	$5.0 \times 10^{-4} \text{ day}^{-1}$	Magnetar spin-down
String sector coupling	SSq	0.57	BH dynamics
Buoyancy coupling	β_i	0.603	Multi-system
SCm completeness	H_{SCm}	≈ 0.99	Heaviside threshold
SCm phonon frequency	ω_{SCm}	$2\pi \times 1.25 \text{ THz}$	Phonon resonance
SCm vacuum density	ρ_{SCm}	$7.09 \times 10^{-37} \text{ J/m}^3$	Fundamental

§SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

Observable	UQFF Prediction	SM / Experiment	Source	Alignment
Phonon frequency	ω_{SCm}	$2\pi \times 1.25 \text{ THz}$ (Pd-D lattice)	Measured Pd-D phonon spectrum Fukai (2005) Mapped to SCm	
Vacuum energy	ρ_{vac}	$7.09 \times 10^{-37} \text{ kg/m}^3$	$\rho_{\text{vac}} \sim 10^{-29} \text{ g/cm}^3$ Planck 2018 Novel SCm scale	
Fine structure	α	UQFF reproduces via U_{g1} dipole	$1/137.036$ PDG 2024 99.9%	

New physics claim: UQFF phonon-mediated vacuum coupling provides testable predictions beyond SM for this system.

Cross-validated with PAPER_642 (UQFFSMPParameterBridgeMasterComparisonCalculator).

§A Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

§A.1 Sector Classification ****Sector:**** SCm-phonon (lattice resonance)

§A.2 Lagrangian Density $\mathcal{L}_{\text{SCm}\backslash\text{phonon}} = \sum_{i=1}^{26} \left[U_{g,i} + U_{m,i} + U_{A,i} - U_{b,i} \right] \cdot S_{26}([SSq]) \cdot$

$\Phi_{1.25\text{THz}}(\omega, \Gamma)$

§A.3 Euler-Lagrange Equation of Motion $\frac{\partial \mathcal{L}}{\partial \phi} - \frac{\partial \mathcal{L}}{\partial \mu} \frac{\partial \mathcal{L}}{\partial \phi} = 0$ implies $F_{U,Bi_i} = -\nabla U_{\text{eff}} + \Phi \cdot S_{26} \cdot E_{\text{net}}$

§A.4 Cosmogenesis Linkage Chain PAPER_877 axioms $\rightarrow$ SCm vacuum $\rightarrow$ phonon ω_{SCm} $\rightarrow$ lattice resonance $\rightarrow$ F_{U,Bi_i} unified force $\rightarrow$ observational prediction

Appendix: Session 225 Cross-References (PAPER_1000–1081)

> Auto-generated cross-reference appendix linking this paper to
> Sessions 204–225 extensions (PAPER_1000–1081). Added by
> update_corpus_crossrefs.py (Session 225, April 2026).

Paper	Title
PAPER_1022	GW Phonon Strain SCm Modulation of $h(t)$
PAPER_1002	AGN Buoyancy-Corrected Eddington Luminosity
PAPER_1009	3C273 AGN F_{U,Bi_i} Jet Modulation
PAPER_1010	TON618 AGN F_{U,Bi_i} Jet Modulation
PAPER_1037	AGN Buoyancy Jet Calculator — SCm Jet Launching
PAPER_1048	M-Sigma Phonon-Corrected Relation
PAPER_1041	SCm Cool-Core Buoyancy Balance AGN Feedback
PAPER_1079	Galaxy Cluster Cooling-Flow Buoyancy Suppression
PAPER_1020	Cosmic Ray Phonon Acceleration DSA Spectrum
PAPER_1021	Pulsar Timing Phonon TOA Residual
PAPER_1023	Neutrino Oscillation Phonon PMNS Matrix SCm
PAPER_1024	Magnetar Giant Flare SCm Phonon Reservoir
PAPER_1072	SCm Activation Function Phonon Threshold
PAPER_1073	SCm Phonon-Driven Inflation Vacuum Buoyancy
PAPER_1069	VDS-DVP-BSH Hybrid Calculator Unified

15 cross-reference(s) identified.

References

- Abbott et al. (LIGO Scientific and Virgo Collaborations, 2016). *Observation of Gravitational Waves from a Binary Black Hole Merger*. Phys. Rev. Lett. **116**, 061102 — arXiv:1602.03837 — doi:10.1103/PhysRevLett.116.061102
- Murphy, D. (2026). *Unified Quantum Field Framework (UQFF): Star-Magic v5.x Whitepaper Series*. Star-Magic Repository — github.com/Daniel8Murphy0007/Star-Magic
- Schmidt, M. (1963). 3C 273: A star-like object with large red-shift. Nature **197**, 1040 — doi:10.1038/1971040a0
- Richards, G.T. et al. (2006). *The Sloan Digital Sky Survey Quasar Survey*. AJS **166**, 470 — arXiv:astro-ph/0601434 — doi:10.1086/506525
- de Vaucouleurs, G. (1948). *Recherches sur les Nebuleuses Extragalactiques*. Ann. Astrophys. **11**, 247

6. Kennicutt, R.C. & Evans, N.J. (2012). *Star Formation in the Milky Way and Nearby Galaxies*. ARA&A **50**, 531 — arXiv:1204.3552 — doi:10.1146/annurev-astro-081811-125610
7. Sofue, Y. & Rubin, V. (2001). *Rotation Curves of Spiral Galaxies*. ARA&A **39**, 137 — arXiv:astro-ph/0010594 — doi:10.1146/annurev.astro.39.1.137
8. Fabian, A.C. (2012). *Observational Evidence of Active Galactic Nuclei Feedback*. ARA&A **50**, 455 — arXiv:1204.4114 — doi:10.1146/annurev-astro-081811-125521
9. McNamara, B.R. & Nulsen, P.E.J. (2007). *Heating Hot Atmospheres with Active Galactic Nuclei*. ARA&A **45**, 117 — arXiv:0709.4098 — doi:10.1146/annurev.astro.45.051806.110625
10. Heckman, T.M. & Best, P.N. (2014). *The Coevolution of Galaxies and Supermassive Black Holes*. ARA&A **52**, 589 — arXiv:1403.4620 — doi:10.1146/annurev-astro-081913-035722
11. Rugh, S.E. & Zinkernagel, H. (2002). *The Quantum Vacuum and the Cosmological Constant Problem*. Stud. Hist. Phil. Mod. Phys. **33**, 663 — arXiv:hep-th/0012253 — doi:10.1016/S1355-2198(02)00033-3
12. Weinberg, S. (1989). *The Cosmological Constant Problem*. Rev. Mod. Phys. **61**, 1 — doi:10.1103/RevModPhys.61.1
13. Blandford, R.D. & Znajek, R.L. (1977). *Electromagnetic extraction of energy from Kerr black holes*. MNRAS **179**, 433 — doi:10.1093/mnras/179.3.433
14. Blandford, R.D. & Payne, D.G. (1982). *Hydromagnetic flows from accretion discs and the production of radio jets*. MNRAS **199**, 883 — doi:10.1093/mnras/199.4.883
15. Ashcroft, N.W. & Mermin, N.D. (1976). *Solid State Physics*. Harcourt
16. Kittel, C. (2004). *Introduction to Solid State Physics*. 8th ed. Wiley
17. Feynman, R.P. (1982). *Simulating Physics with Computers*. Int. J. Theor. Phys. **21**, 467 — doi:10.1007/BF02650179